

PSYC 4141, Week 2 2026

Section A

Learning outcomes

- By the end of the lecture you should be able to:
 - Understand distinctions between types of data
 - Be aware of some key notations for discussing numbers
 - Be aware of measures of central tendency

Types of Data

- Broadly speaking, there are two types of data
- The first type is called 'nominal' data
- Nominal variables are those in which a case falls into one of two or more categories
- Examples are eye colour, socio-economic status, marital status and nationality.
 - Nominal variables are also referred to as categorical or qualitative variables

Types of Data (Continued)

- The second type of data is called 'numerical' data
- Numerical variables are those in which a case is assigned a numerical value
- Examples are age, height, weight, IQ, test scores, income, distance or temperature
- Numerical variables are also referred to as score or quantitative variables

Nominal or Numerical?

- Which type of data are each of the following?
 - Facebook user (i.e. yes or no)
 - Use of Facebook (i.e. hours per day)
 - Network (i.e. number of friends on Facebook)
 - How do you access Facebook (i.e. mobile phone, PC or laptop)
 - Use of Facebook (i.e. Less than five hours per day or at least five hours per day)

Scales of Measurement

- Four different scales of measurement exist:
 - Nominal categorisation – Placing cases into named categories (e.g. high jumpers could be categorised based on their nationality)
 - Ordinal categorisation – This ranks cases based on their order on a given variable (i.e. high jumpers can be ranked 1st, 2nd, 3rd etc.)
 - Interval categorisation – Where the distances between the sequential points on the scale are equal (e.g. the temperature at the time of the event)
 - Ratio categorisation – The same as interval categorisation, but with an absolute zero (e.g. height jumped successfully)

Functions of Statistical Techniques

- Statistical techniques perform three key functions:
 - Descriptive statistics, as you would guess, is used to describe the information collected
 - Inferential statistics relates to the confidence with which we can generalise from our sample to the population of interest
 - Data reduction techniques allow a researcher to make sense of large amounts of data through using more advanced statistics

Interim Conclusions

- Data comes in two broad forms
 - Nominal variables are those in which cases are placed in categories
 - Numerical variables concern those in which cases are given a score value
- It is important to understand which of these types of data each of your variables are as this will influence your analysis
- The measurement of a variable can take four forms: nominal, ordinal, interval and ratio
- Statistics can be used to describe, make inferences about or reduce your data

Notation

- Scores for a particular variable are generally represented by the letter X
- Scores for two variables are generally represented by X and Y
- The number of scores in a distribution is represented by N .
- $\sum X$ (“sigma X ”) is the sum of X scores
- M refers to the arithmetic mean of the scores



Central tendency

- The mean is the average from a set of scores

$$M = \sum X / N$$

The mean

- What is the mean of 7,8,8,7,3,1,6,9,3,8?
 - $\Sigma X = 7+8+8+7+3+1+6+9+3+8 = 60$
 - $N = 10$
 - $\text{Mean} = 60/10 = 6$

The median

- The middle score when all scores are arranged from lowest to highest
- Median of 7,8,8,7,3,1,6,9,3,8

– 1 3 3 6 7 7 8 8 8 9



median

- Median is the average (mean) of the 5th and 6th scores, so the median is 7

The mode

- Most common single number in a distribution
- Mode of 7,8,8,7,3,1,6,9,3,8
 - 1 3 3 6 7 7 8 8 8 9
 - Mode is 8

Thinking about central tendency

- “Of the three, which measure of central tendency should be used?”

Why do we need *three* measures?”

- Examples where the mean isn't possible
- Examples where the mean isn't meaningful
- Questions without a simple answer
- Examples of where some measures may be misleading or controversial

Using statistics carefully

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Fighting fraud



Summary

- Psychology involves using data in different ways
 - This can make a big difference you how we handle data
 - We need to understand the scale of data
- Central tendency describes representative values
 - Different types of central tendency measure
 - Ask yourself which is better in the circumstances

PSYC 4141, Week 2

Section B

Evidence in psychology

- "In listening to stories we tend to suspend disbelief in order to be entertained, whereas in evaluating statistics we generally have an opposite inclination to suspend belief in order not to be beguiled."
 - John Allen Paulos, in Stories vs. statistics, New York Times, 24 October 2010

Central tendency

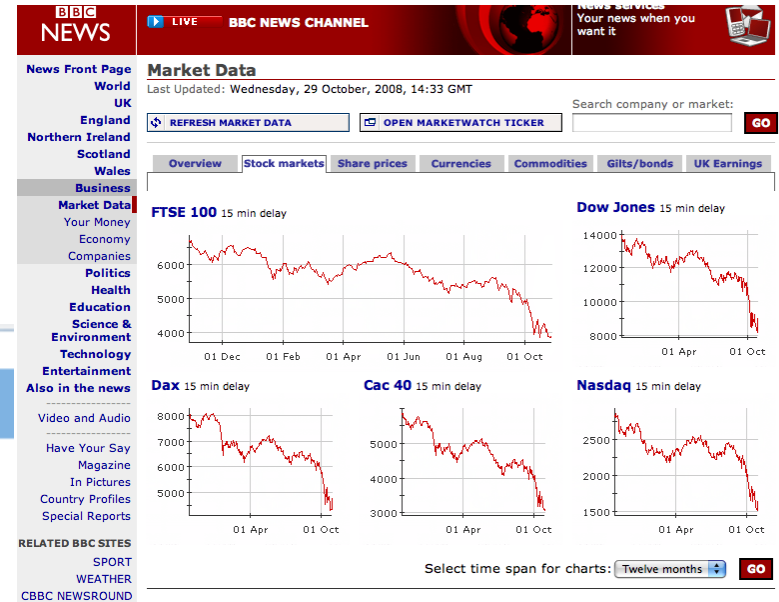
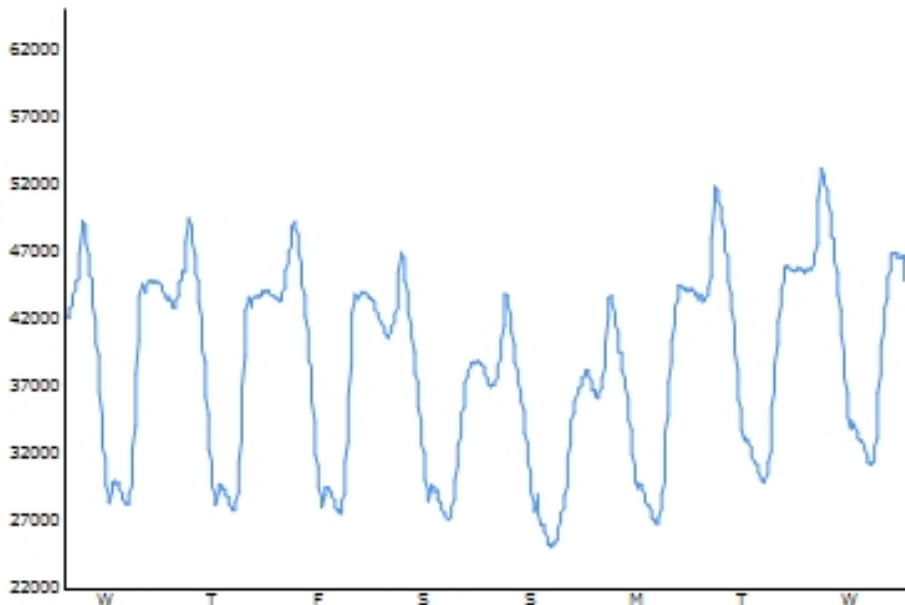
- Mean, median and mode
- Each of them can be useful
- Each describe something potentially meaningful about a sample or a population
- On their own, they don't tell the whole story...

The importance of variability

- What questions might you ask or think about in deciding on your application to a University to study Psychology in 2023?
 - What is the average cost of living in place X?
 - What is the most common degree outcome that students get?
 - How long does it take for students to find a graduate-level job when they leave?
 - On average, how many hours of teaching is there in the 1st year?

Illustrations of variability

Electricity demand - Last 8 Days



Expressing variability

- Visual depiction of variability
 - The amount of spread
 - The shape of spread
- Numerical information about variable

Ways to assess variability

- *Range* as a measure of variability
- The inter-quartile range and semi-inter-quartile range.
- Both of these are affected by extreme scores
- Deviation is the distance from the mean: $X - M$
 - The mean deviation must be always be zero!
 - Therefore, we square the deviations
 - Variance is the mean squared deviation
 - Standard deviation = $\sqrt{\text{variance}}$

Variability (in words)

- Variance tells you how spread out the scores are around the mean
- Standard deviation is (roughly) the average amount that scores differ from the mean

Calculating the standard deviation

- Find the standard deviation for the scores:
- 8, 6, 6, 9, 6, 5, 6, 2

Calculating the standard deviation

Scores	Deviation	Deviation ²	
8	2	4	$\frac{\sum(X - M)^2}{N}$
6	0	0	
6	0	0	
9	3	9	
6	0	0	$SD = \sqrt{\frac{\sum(X - M)^2}{N}}$
5	-1	1	
6	0	0	
2	-4	16	1.94
48	Mean = 6	SS = 30	

Example data

Q2 - What is your age?

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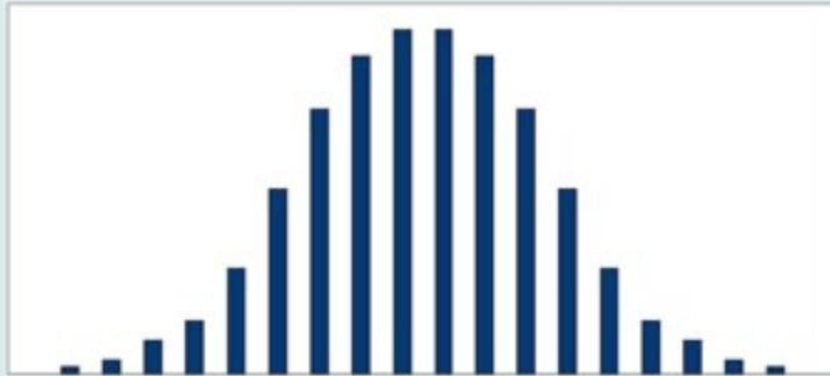
#	Field	Minimum	Maximum	Mean	Std Deviation	Variance	Count
1	What is your age?	18.00	30.00	18.61	1.39	1.94	153

Q27 - How many friends do you have on Facebook?

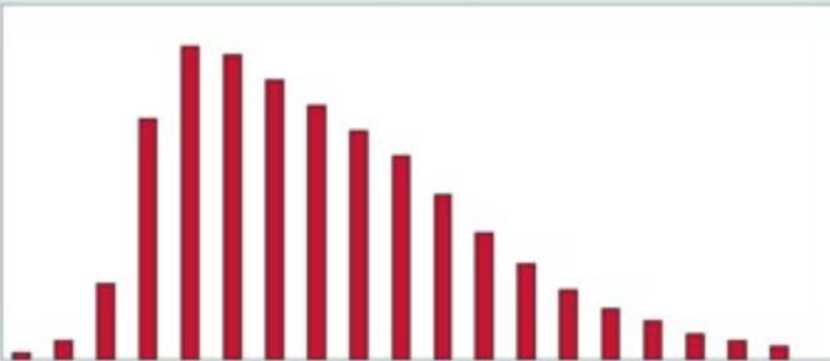
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#	Field	Minimum	Maximum	Mean	Std Deviation	Variance	Count
1	How many friends do you have on Facebook?	0.00	2510.00	207.80	333.63	111306.97	84

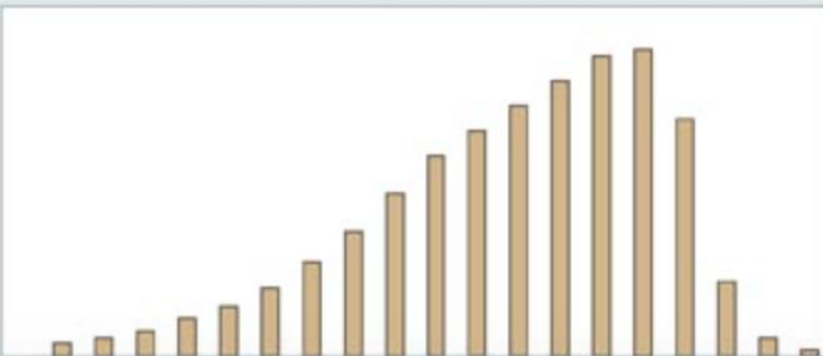
Normal distribution



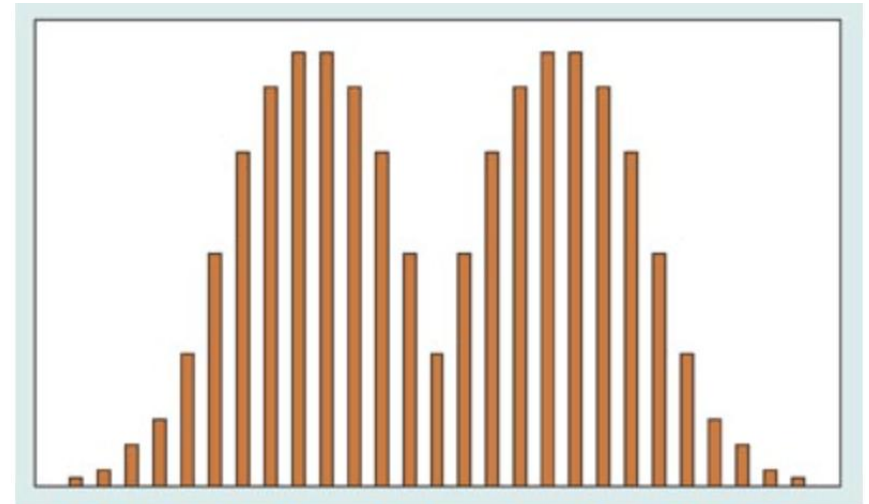
Positive skew



Negative skew



Variance and distributions



Interpreting the standard deviation

- Baron-Cohen, Leslie & Frith (1985)
 - Studied whether children with autism could distinguish between knowledge and false belief
 - Tested 20 children with autism, 14 children with Down's syndrome and 14 typically-developing children

Table 1. *Means, SDs and ranges of Chronological Age (CA) and Mental Age (MA) in years; months*

Diagnostic groups	<i>n</i>		CA	Nonverbal* MA	Verbal** MA
Autistic	20	Mean	11;11	9;3	5;5
		SD	3;0	2;2	1;6
Down's syndrome	14	Mean	10;11	5;11	2;11
		SD	4;1	0;11	0;7
Normal	27	Mean	4;5	–	–
		SD	0;7		

*Leiter International Performance Scale.

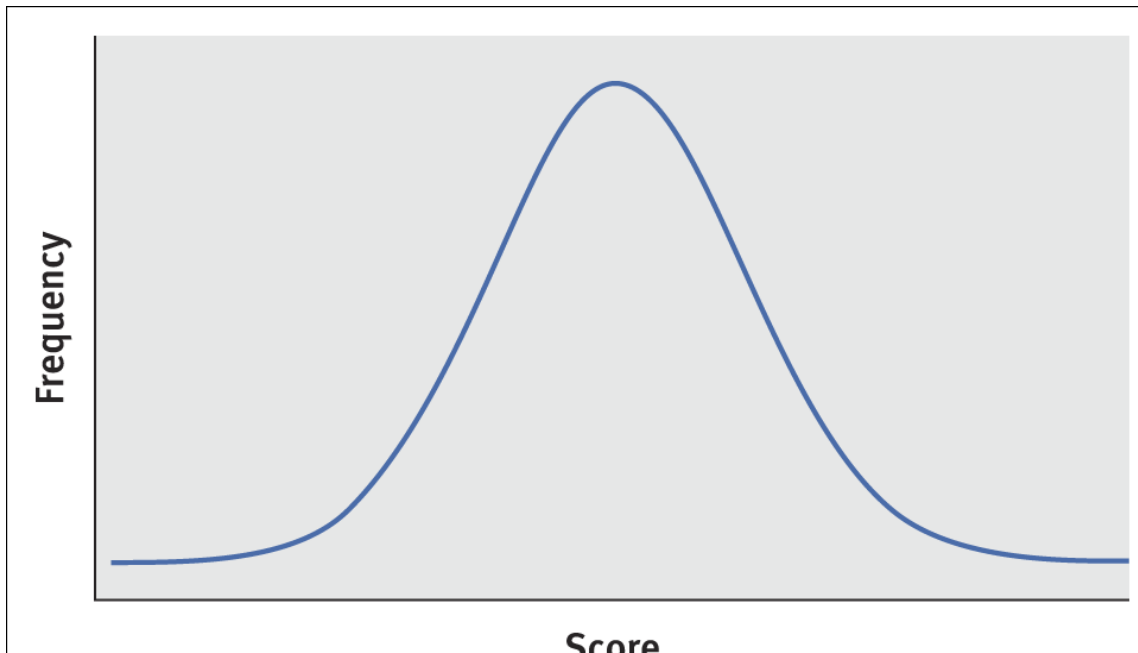
**British Picture Vocabulary Test

Summary

- Central tendency describes representative values
 - Different types of central tendency measure
 - Ask yourself which is better in the circumstances
- Variability describes the spread of scores
 - Variability is important
 - It can be calculated in different ways
 - 'Standard deviation' is a vital measure of variability

Properties of the normal curve

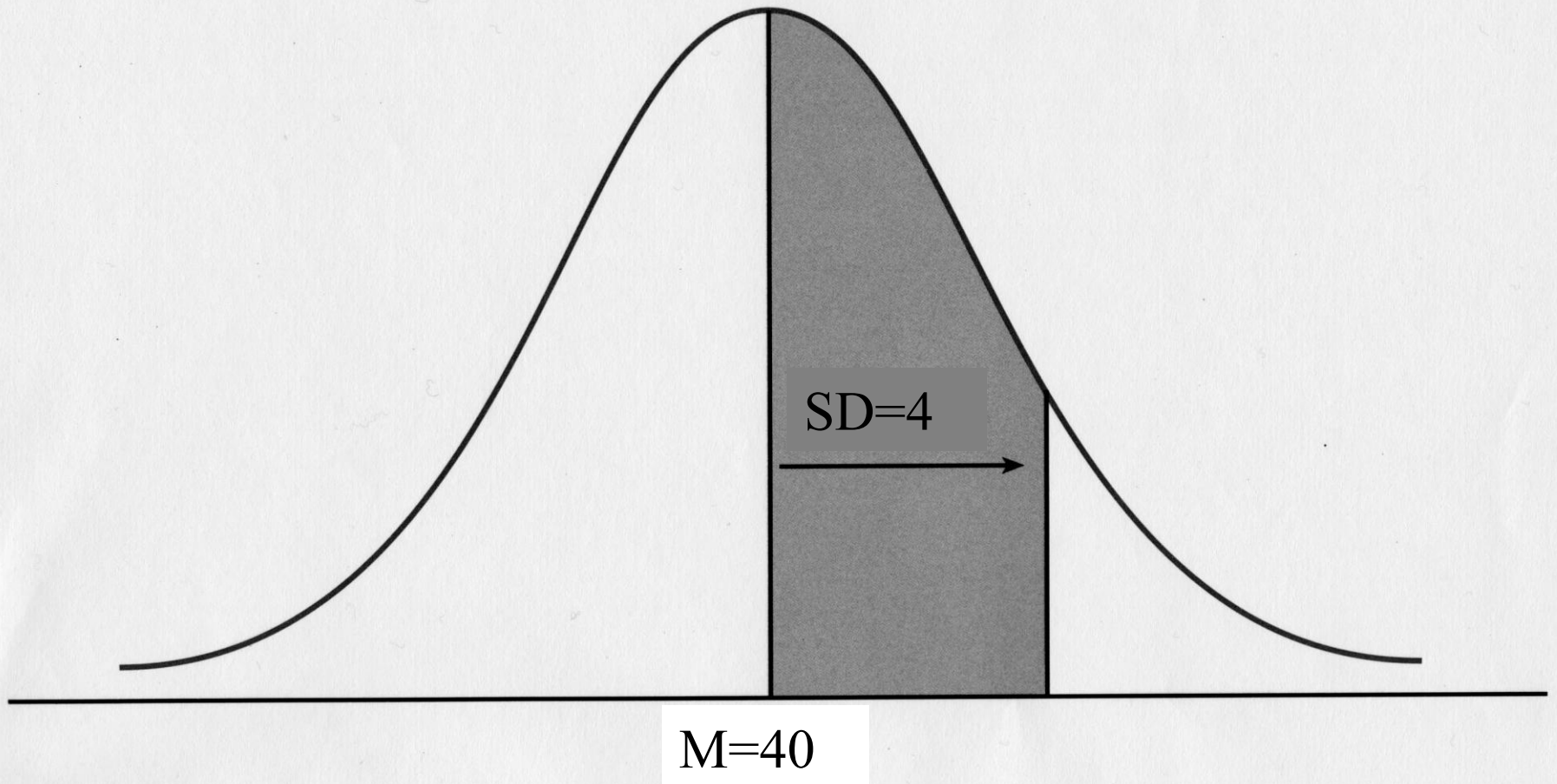
- When we examine variability, as a distribution of events (scores, frequencies etc.) we often find that we produce a normal distribution of scores Normal distribution curves are bell-shaped
 - They are bilaterally symmetrical
 - The tails of the curve approach, but do not touch, the X-axis.
 - Although the graph will go on indefinitely, the area under the graph is considered to have a unit of 1.00
 - The mean, median, and mode are the same value
 - (You can estimate the mode and median by simply looking at a graph: the mode is the value with the highest frequency, and the median is the middle point. It is harder to estimate the mean, however, as that depends on the range of values. Unless it is a normal distribution!)



The importance of the normal distribution curve

- Many of the variables in behavioural sciences closely approximate the normal curve.
- Many inference tests require sampling distributions that are normally distributed.

Properties of the normal distribution



Summary

- We need to be comfortable with notions of central tendency and variability to understand many psychological phenomena
- The normal distribution is important for thinking about how to interpret descriptive data